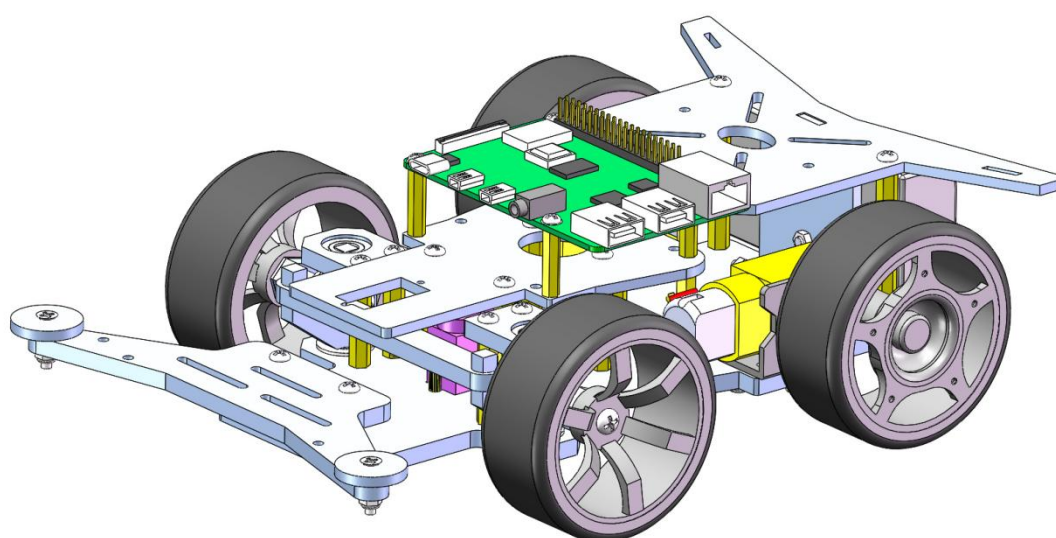


How to Assemble the Real-Wheel Drive Steering Car Chassis for Raspberry Pi

Table

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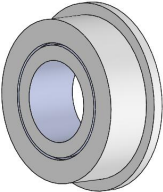
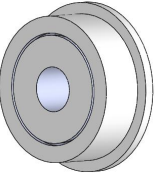
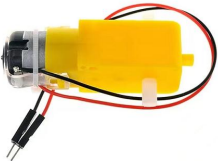
1. Parts List

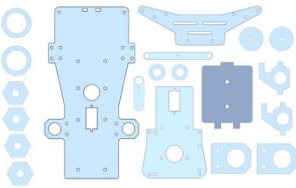
1.1 Connecting/Fixing Parts

Picture	Name	Description	Qty
	self-tapping screw	P1.2*5mm self-tapping screws	6
	screw	M2*8mm round screw	10
	screw	M2*12mm round screw	2
	screw	M3*10mm round screw	30
	screw	M3*25mm round screw	5
	screw	M3*8mm countersunk head screw	3

	screw	M3*14mm countersunk head screw	3
	screw	M3*8mm round screw	14
	screw	M4*40mm round head screw	4
	screw	M2.5*8mm round screw	9
	nut	M3 nut	8
	nut	M2 self-locking nut	2
	nut	M3 self-locking nut	8

	nut	M4 self-locking nut	4
	copper column	M2*10mm double through knurled copper post	4
	copper column	M2.5*20mm double pass copper column	4
	copper column	M3*20mm double pass copper column	4
	copper column	M3*10mm double pass copper column	3
	copper column	M3*25mm double pass copper column	6
	copper column	M3*30mm double pass copper column	4

	copper column	M3*45mm double pass copper column	4
	bearing	F687ZZ Flange bearing (standard grade)	4
	bearing	F624ZZ Flange bearing (standard grade)	4
	servo	MG90S servo (180 degrees)	1
	battery case	18650 battery case	1
	motor	TT motor-dual axis reduction ratio 1:48	2
	bracket	TT motor metal bracket	2

	wheel	TT motor wheel	2
	wheel	Hexagonal wheel	2
	acrylic structural parts	Acrylic structural parts of rear-wheel drive front-wheel steering chassis	1

1.2 Tool

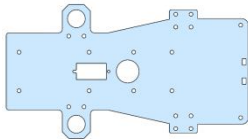
Picture	Name	Description	Qty
	wrench	M3 Simple Wrench	1
	screwdriver	M3 Phillips screwdriver	1
	screwdriver	M1.5 Phillips screwdriver	1

	sleeve	M2/M2.5/M3/M4 Cross sleeve	1
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2. Assemble and Install

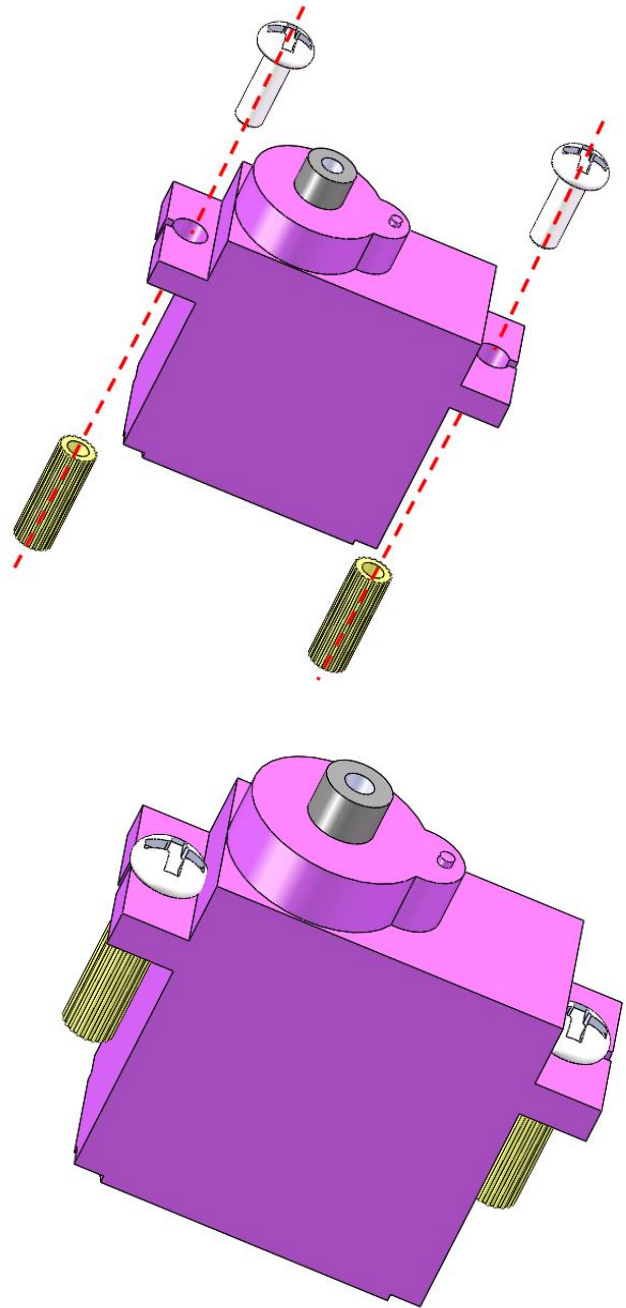
2.1 Assembly steps

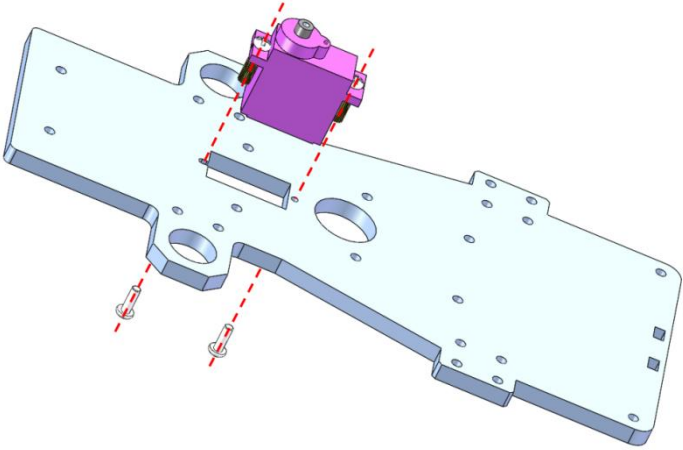
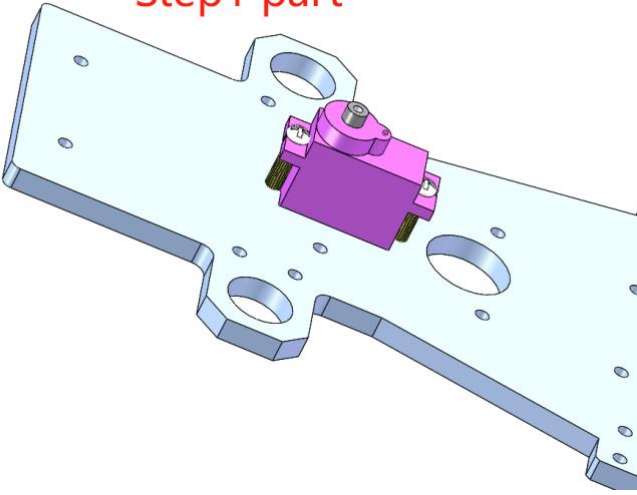
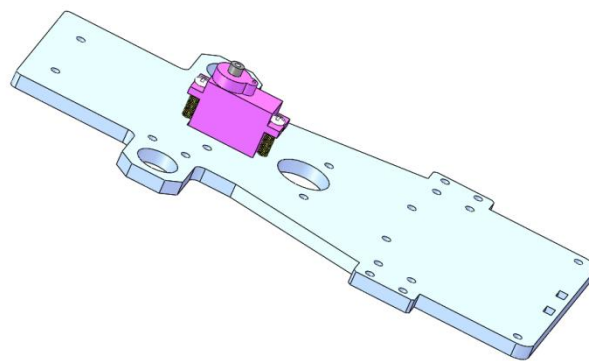
Note: 1. Before assembly, we need to use a screwdriver to peel off the protective paper of the acrylic structure;

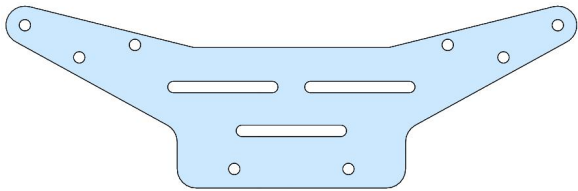
Step	1	Assemble the MG90S servo			Tool	M3 Phillips screwdriver
						
List	Part Name	Quantity	Unit	Picture		
	acrylic chassis framee	1	PCS			
	MG90S servo	1	PCS			
	M2*8 mm round head screw	4	PCS			
	M2*10 mm double through knurled copper column	2	PCS			
	Description			Demo		

A

1. Use 2pcs M2*8mm round head screw to fix M2*10mm double through knurled copper column on MG90S servo;

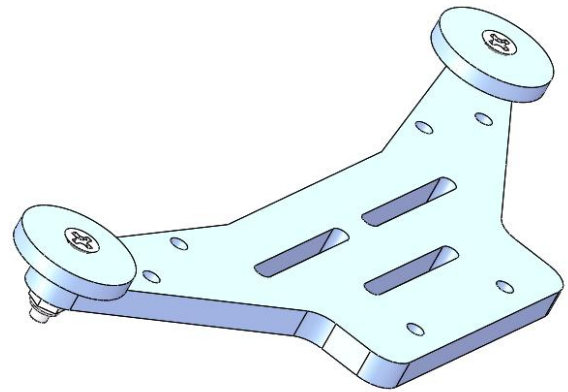
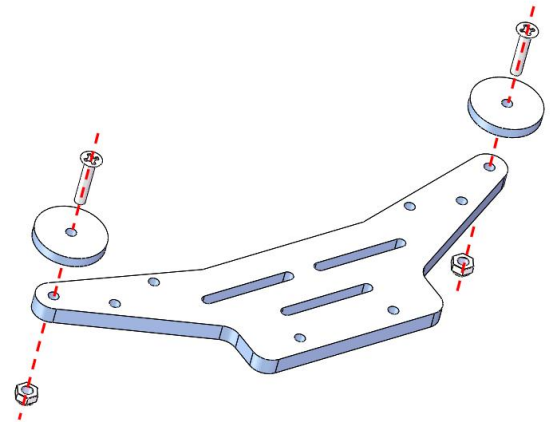


	2. Use 2pcs M2*8mm round head screws to install the semi-finished product completed in the previous step on the acrylic chassis framee; Note: servo installation direction			 Step1 part 	
Step	2	Assemble the front acrylic expansion plate		Tool	M3 Phillips screwdriver 
List	Part Name	Quantity	Unit	Picture	
	Step 1 part	1	PCS		

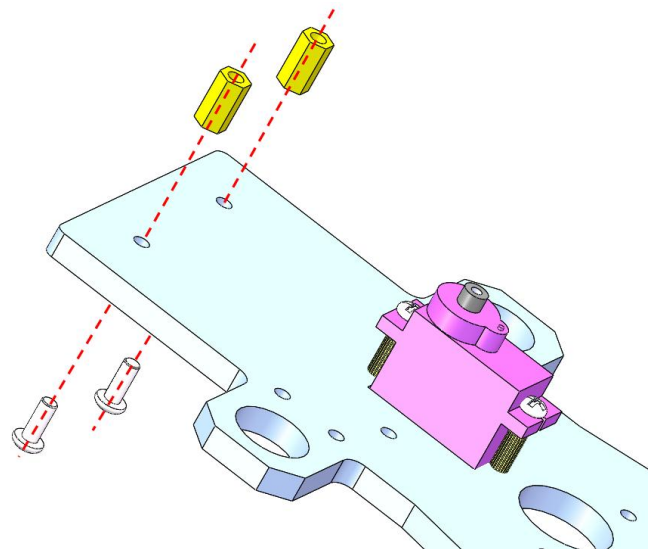
	Front acrylic expansion plate	1	PCS	
	Guide disc	2	PCS	
	M3*14mm countersunk screw	2	PCS	
	M3 self-locking nut	2	PCS	
	M3*10mm double through copper column	2	PCS	
	M3*8mm round head screw	4	PCS	
	Description			Demo

A

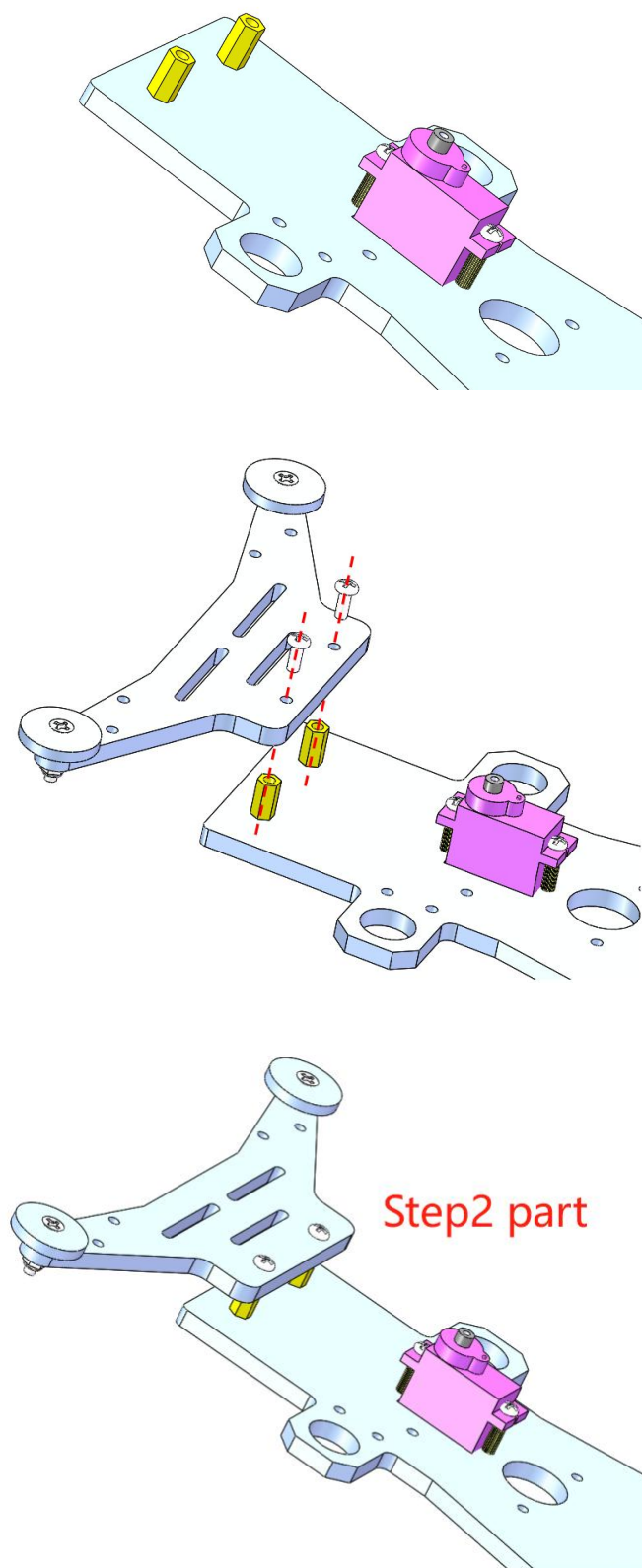
1. Use 2pcs M3*14mm countersunk screws and 2pcs M3 self-locking nuts to fix the guide disc on the front acrylic expansion plate (the self-locking screw does not need to be tightened to ensure that the guide disc can rotate manually);



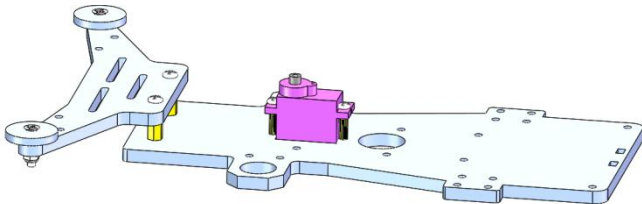
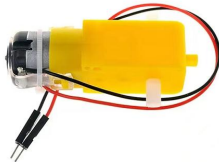
2. Use 2pcs M3*8mm round head screws to fix 2pcs M3*10mm double through copper columns on the acrylic chassis frame;

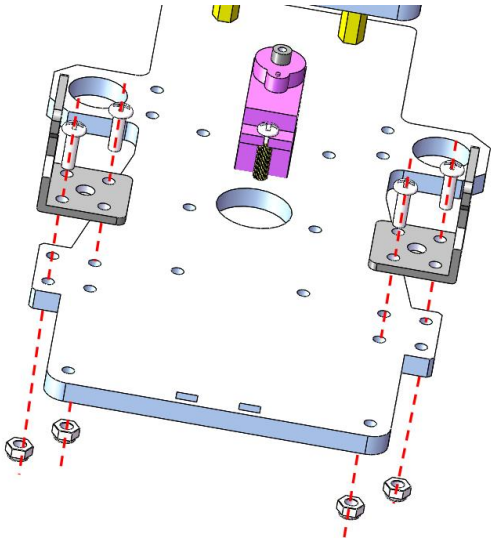


3. Use 2pcs M3*8mm round head screws to fix the front acrylic expansion plate on M3*10mm double copper column;

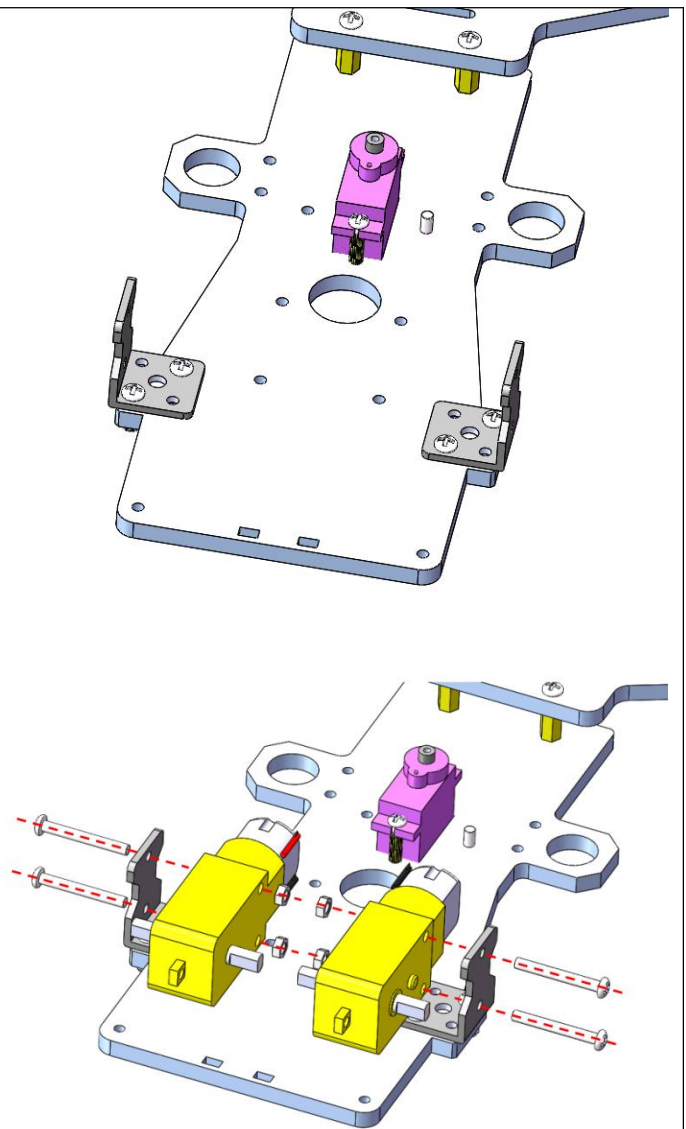


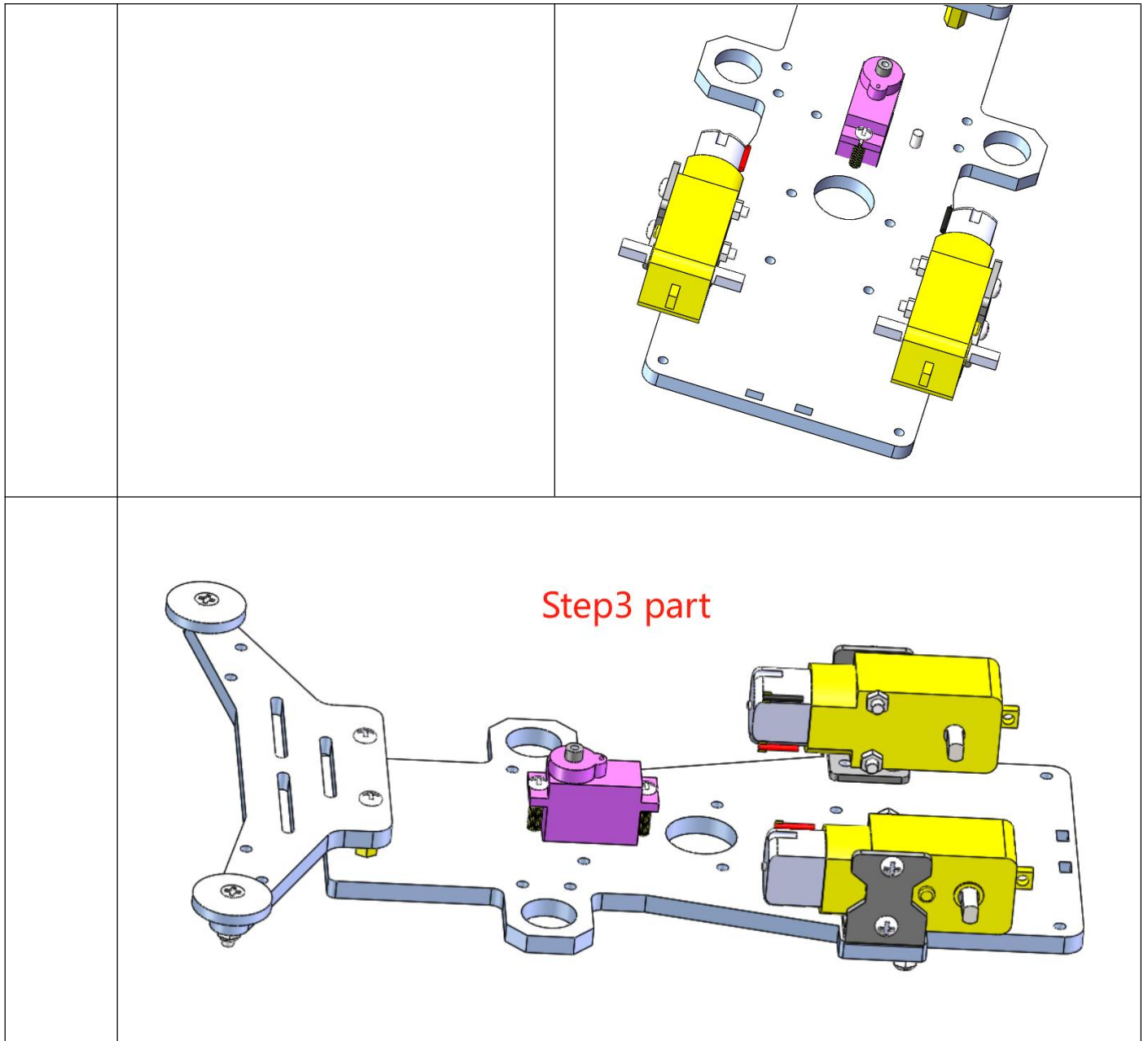
Step	3		TT motor installation	Tool	M3 Phillips screwdriver/
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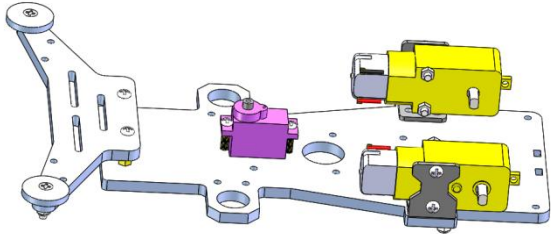
					wrench/cross sleeve
					  
List	Part Name	Quantity	Unit	Picture	
	Step 2 part	1	PCS		
	TT motor bracket	2	PCS		
	TT motor	2	PCS		
	M3*25mm round head screw	4	PCS		
	M3 nut	4	PCS		

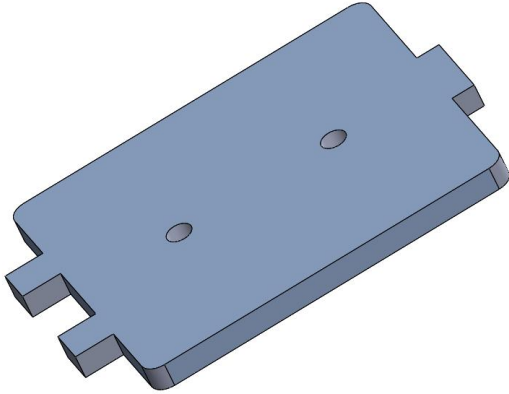
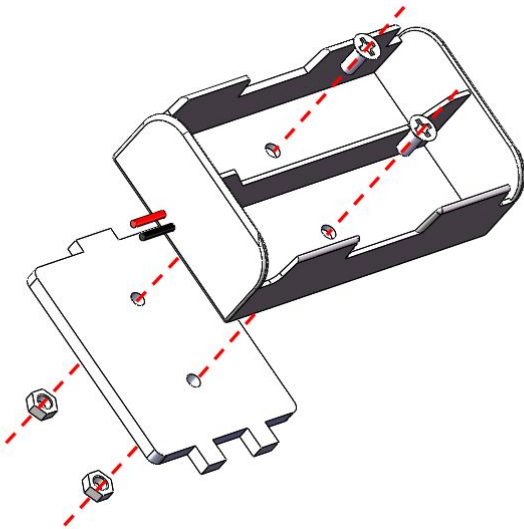
	M3*10mm round head screw	4	PCS	
	M3 self-locking nut	4	PCS	
	Description			Demo
A	1. Use 4 pcs M3*10mm round head screws and 4 pcs M3 self-locking nuts to fix the TT motor brackets on the acrylic chassis frame (pay attention to the installation direction of the TT motor bracket);			

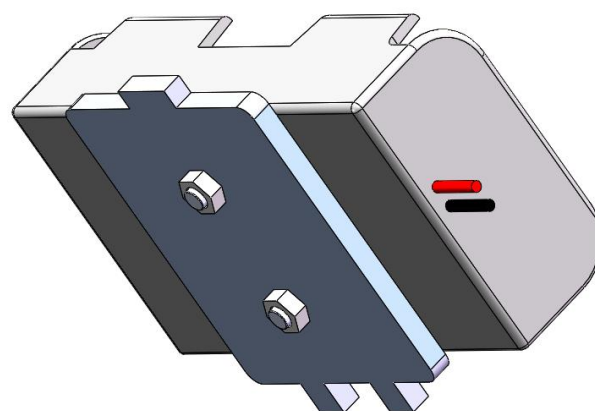
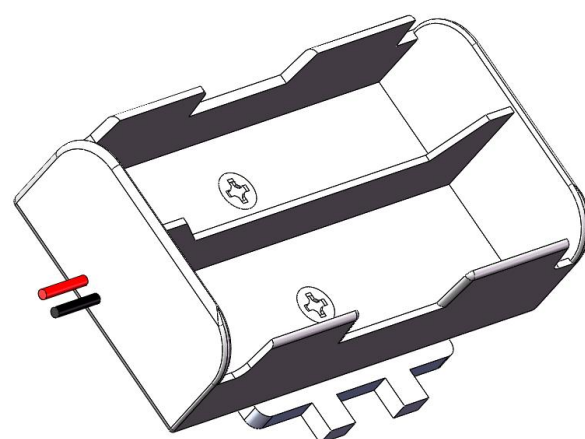
2. Use 4 pcs M3*25mm round head screws and 4 pcs M3 nuts to install the TT motor on the TT motor bracket (note the installation direction of the TT motor);



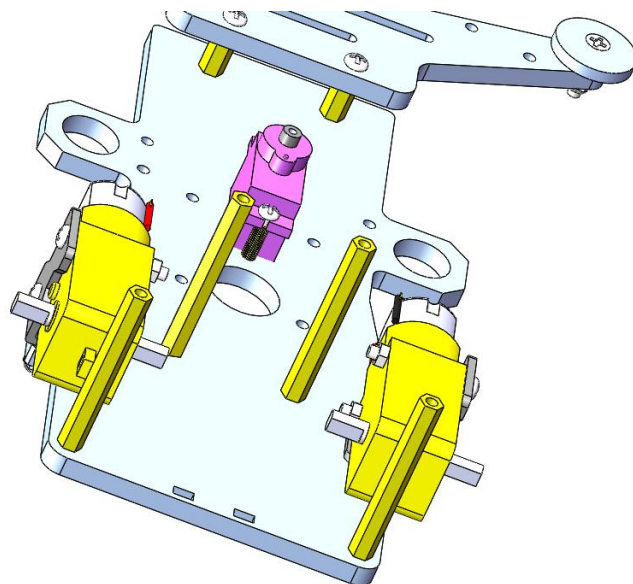
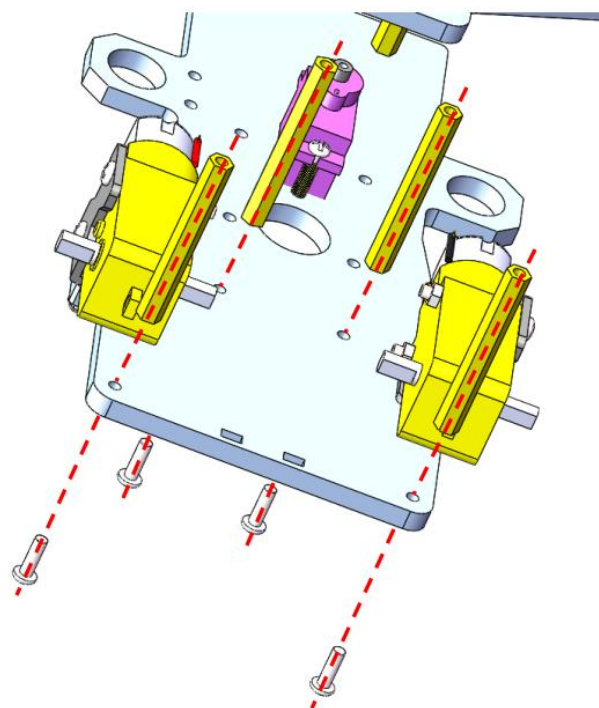


Step	4	Install battery case and back acrylic expansion plate		Tool	M3 Phillips screwdriver
					
List	Part Name	Quantity	Unit	Picture	
	Step 3 part	1	PCS		
	M3*45 mm copper column	4	PCS		
	Battery case	1	PCS		
	M3*8mm countersunk screw	2	PCS		
	M3 nut	2	PCS		

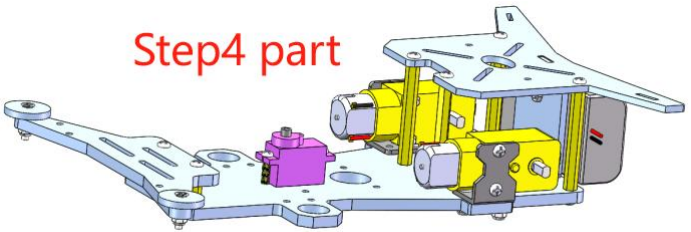
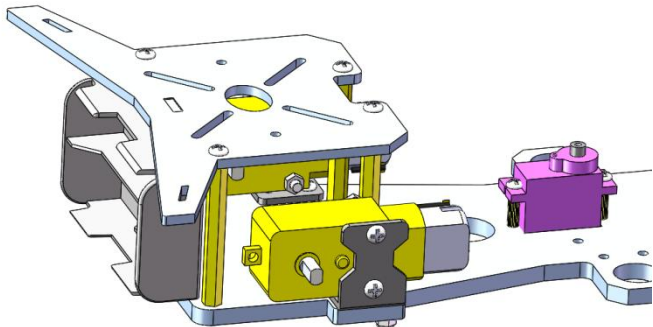
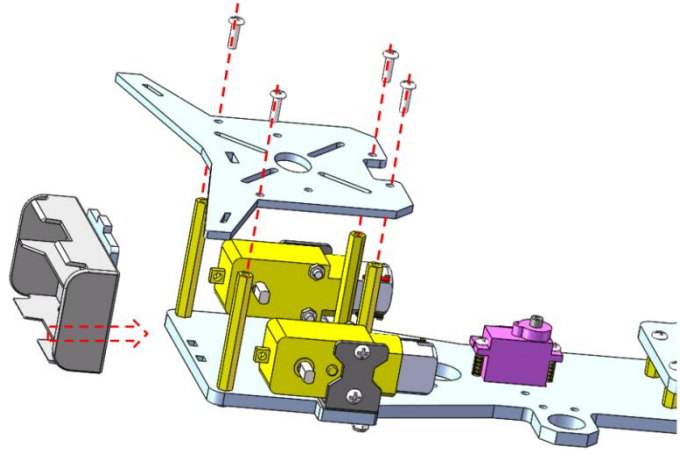
	Battery case base	1	PCS	
	M3*10mm round head screw	8	PCS	
	Description			Demo
A	1. Use 2 pcs M3*8mm countersunk screws and 2 pcs M3 nuts to fix the battery case on the battery case base; Note: The installation direction of the battery case is distinguished by the power cord of the battery case;			



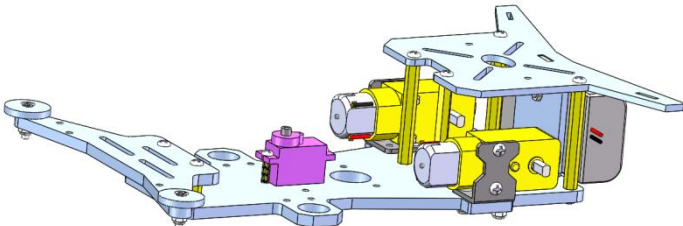
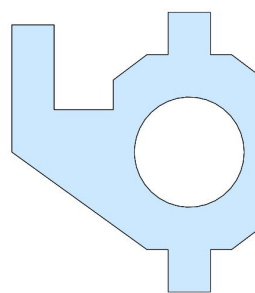
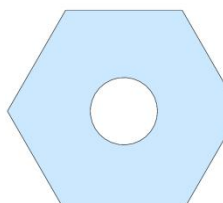
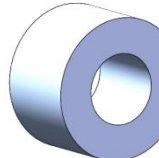
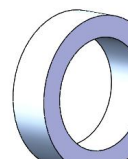
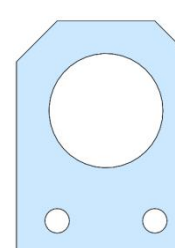
2. Use 4pcs M3*10mm round head screws to fix 4 pcs M3*45mm double through copper columns on the acrylic chassis frame;

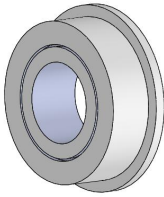
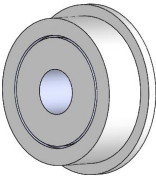


3. The battery case base plate with the battery case is installed on one side of the acrylic chassis frame, and the other side is fixed to the back acrylic expansion plate. The back acrylic expansion plate is fixed to the 4 pcs M3*45mm double-through copper columns by 4 pcs M3*10 mm round head screws;



Step	5	Install the front wheels	Tool	M3 Phillips screwdriver/cross sleeve
				 
List	Part Name	Quantity	Unit	Picture

	Step 4 part	1	PCS	
	front wheel Y-shaped connecting shaft	2	PCS	
	Structural B	4	PCS	
	Structural C	4	PCS	
	Structural D	2	PCS	
	front wheel fixing bracket	2	PCS	

	F687ZZ Flange bearing (standard grade)	4	PCS	
	F624ZZ Flange bearing (standard grade)	4	PCS	
	Hexagonal wheel	2	PCS	
	M4*40mm round head screw	2	PCS	
	M4 self-locking nut	2	PCS	
	M3*10mm round screw	8	PCS	
	M3*25mm double copper column	4	PCS	
	Description			Demo

A

1. Left front wheel installation:

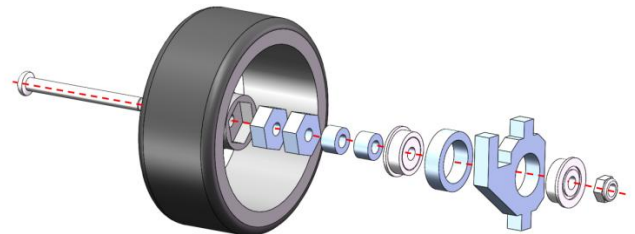
Use 1 pcs M4*40mm round head screw and 1 pcs M4 self-locking nut to fix the hexagonal wheel, 2 structural parts B, 2 structural parts C, F624ZZ flange bearing, structural parts D, front wheel Y-shaped connecting shaft and F624ZZ flange bearing in sequence;

Note:

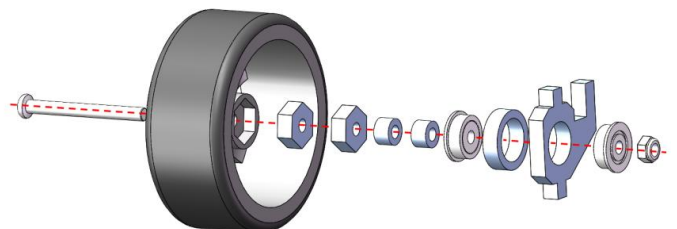
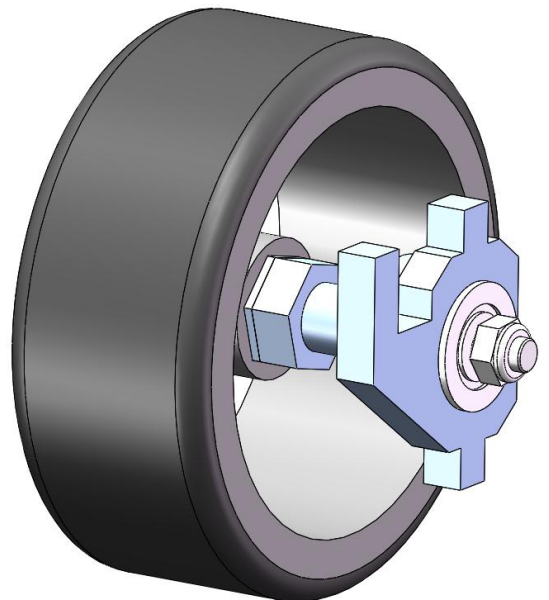
- a. The installation direction of structural part D must be shown in Figure1;
- b. M4 self-locking nut should not be locked to ensure smooth manual rotation of hexagonal wheel;

2. Right front wheel installation:

Use 1 pcsM4*40mm round head screw and 1pcs M4 self-locking nut to fix the hexagonal wheel, 2 structural parts B, 2 structural parts C, F624ZZ flange bearing, structural parts D, front wheel Y-shaped connecting shaft and F624ZZ flange bearing in sequence;



Installation diagram of the left front wheel (Figure 1)



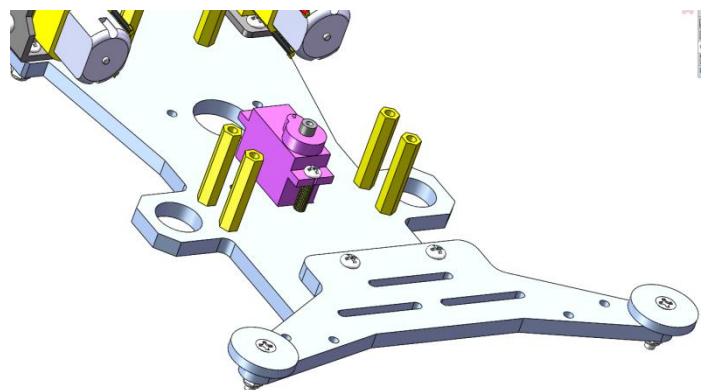
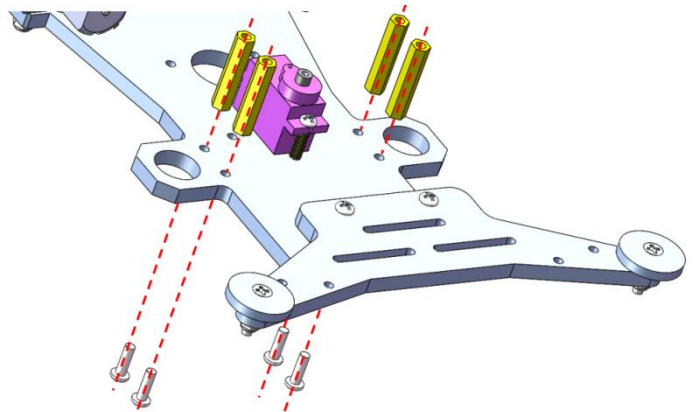
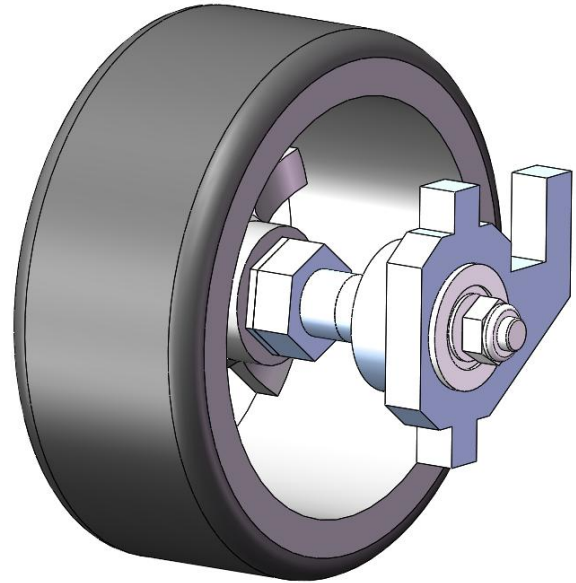
Installation diagram of the right front wheel (Figure 2)

Note:

a. The installation direction of structural part D must be shown in Figure 2;

b.M4 self-locking nut should not be locked to ensure smooth manual rotation of the hexagonal wheel;

3. Use 4 pcs M3*10mm round head screws to fix 4 pcs M3*25mm double through copper columns on the acrylic chassis frame;



4. Installation of the left front wheel to the acrylic chassis frame (Figure 3):

4.1 Install the F687ZZ bearing on the left side front wheel fixing bracket of the acrylic chassis frame;

4.2 Install one end of the front wheel Y-shaped connecting shaft on the left front wheel to the F687ZZ bearing on the acrylic chassis frame, and install the other end of the front wheel fixing bracket in turn with the F687ZZ bearing;

4.3 Use 2 pcs M3*10mm round head screws to install the front wheel fixing bracket on the M3*25mm double through copper columns;

Note: Installation direction of the 2 F687ZZ bearing;

5. Installation of the right front wheel to the acrylic chassis frame (Figure 4):

5.1 Install the F687ZZ bearing on the right side of the acrylic chassis

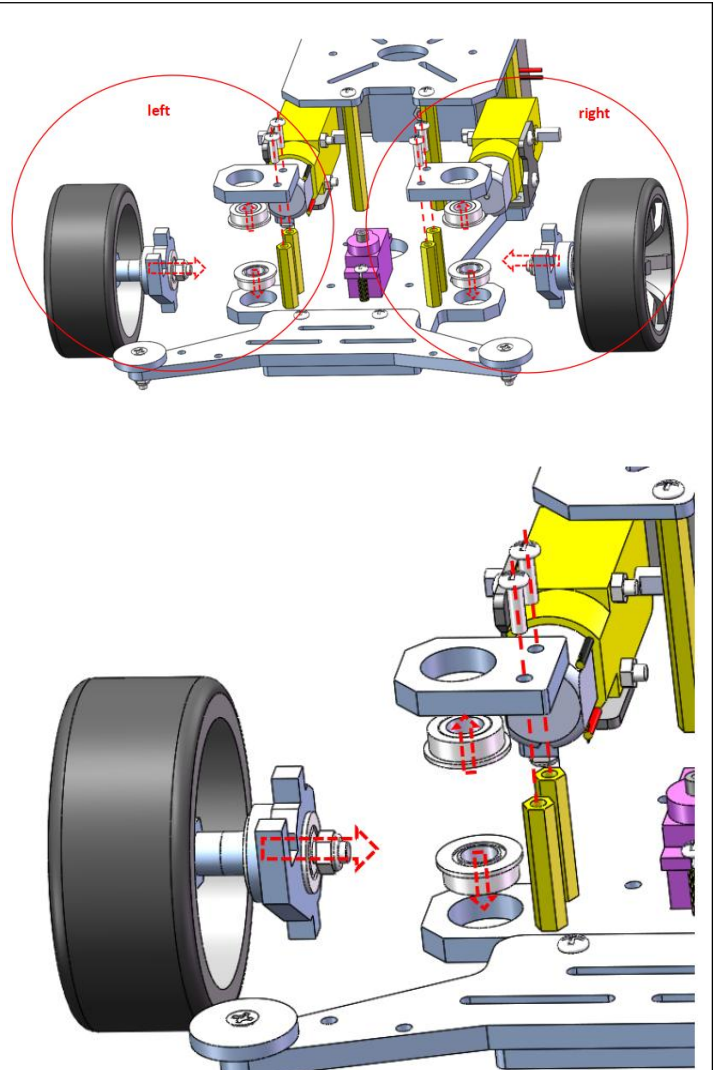


Figure 3

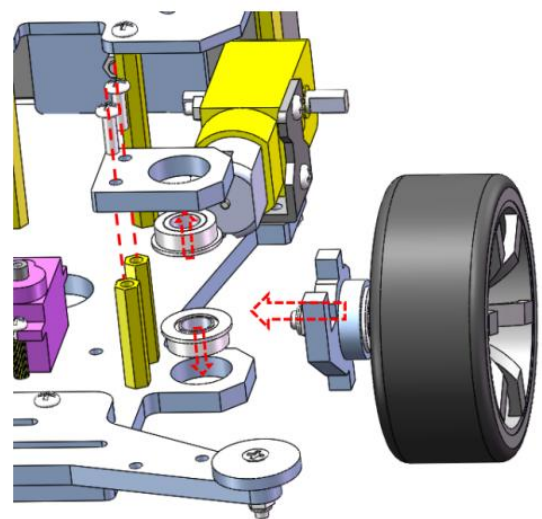


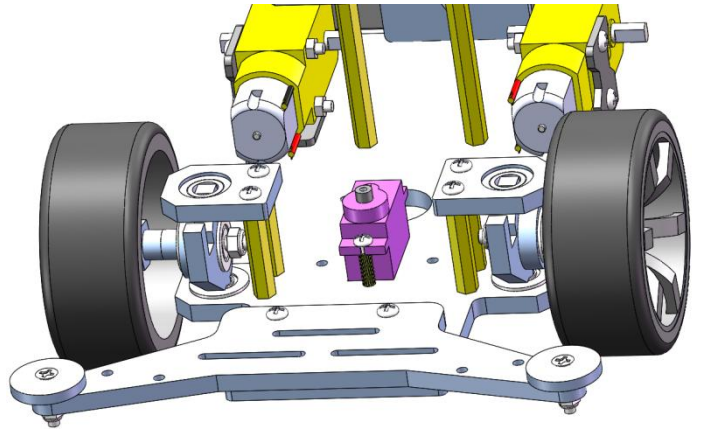
Figure 4

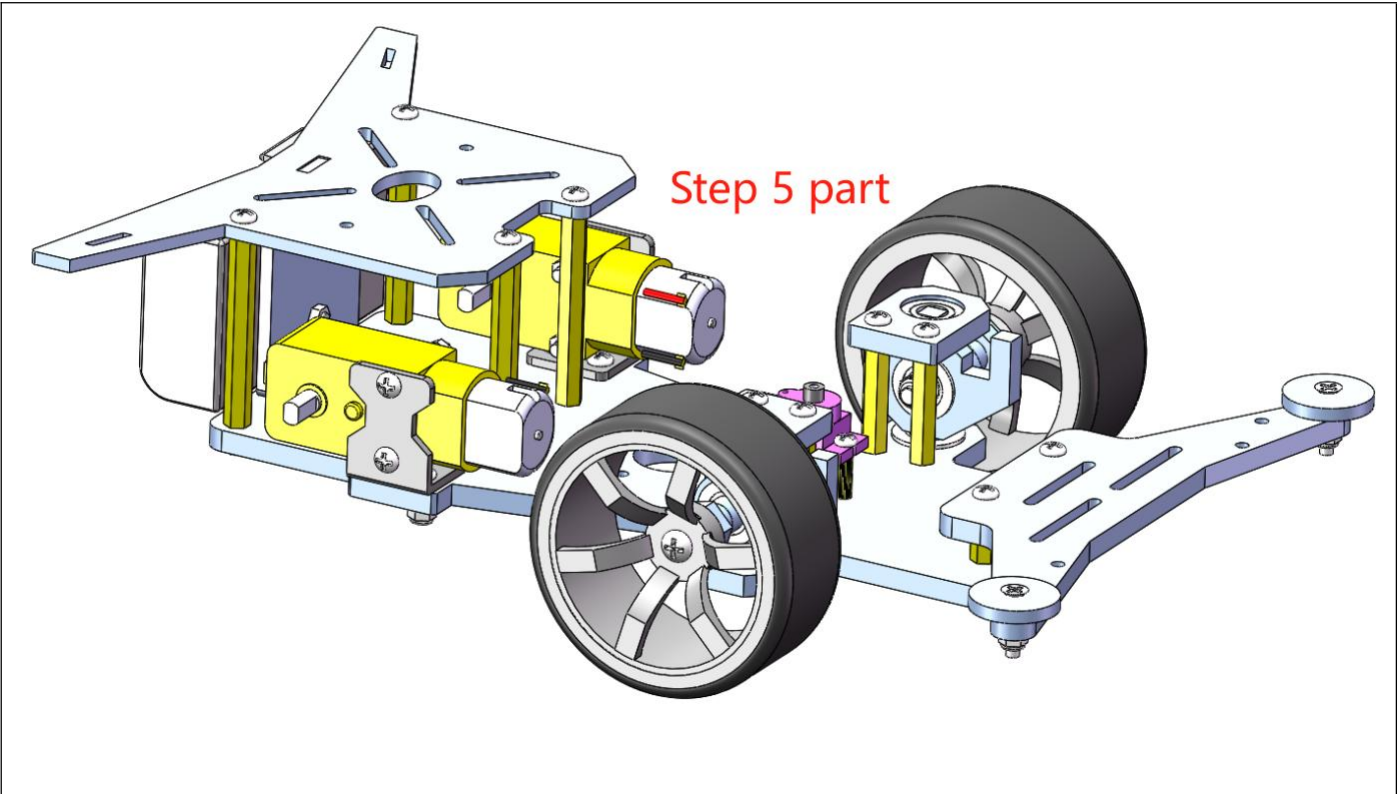
frame;

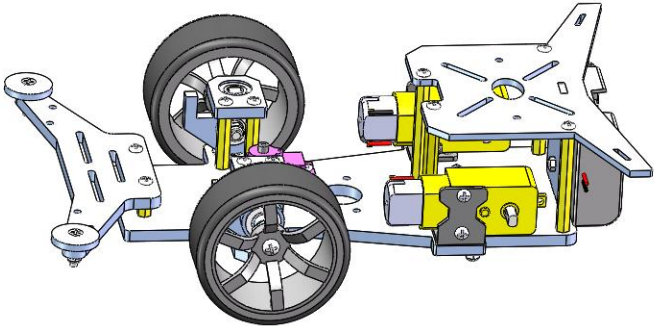
5.2 Install one end of the front wheel Y-shaped connecting shaft on the right front wheel to the F687ZZ bearing on the acrylic chassis frame, and install the other end of the front wheel fixing bracket in turn with the F687ZZ bearing;

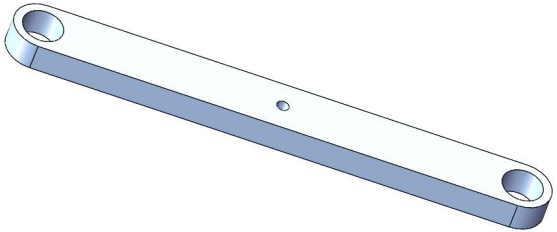
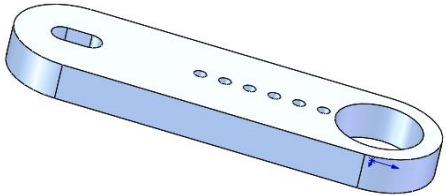
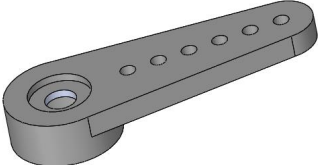
4.3 Use 2pcs M3*10mm round head screws to install front wheel fixing bracket on M3*25mm double through copper column;

Note: Installation direction of the 2 F687ZZ bearing;



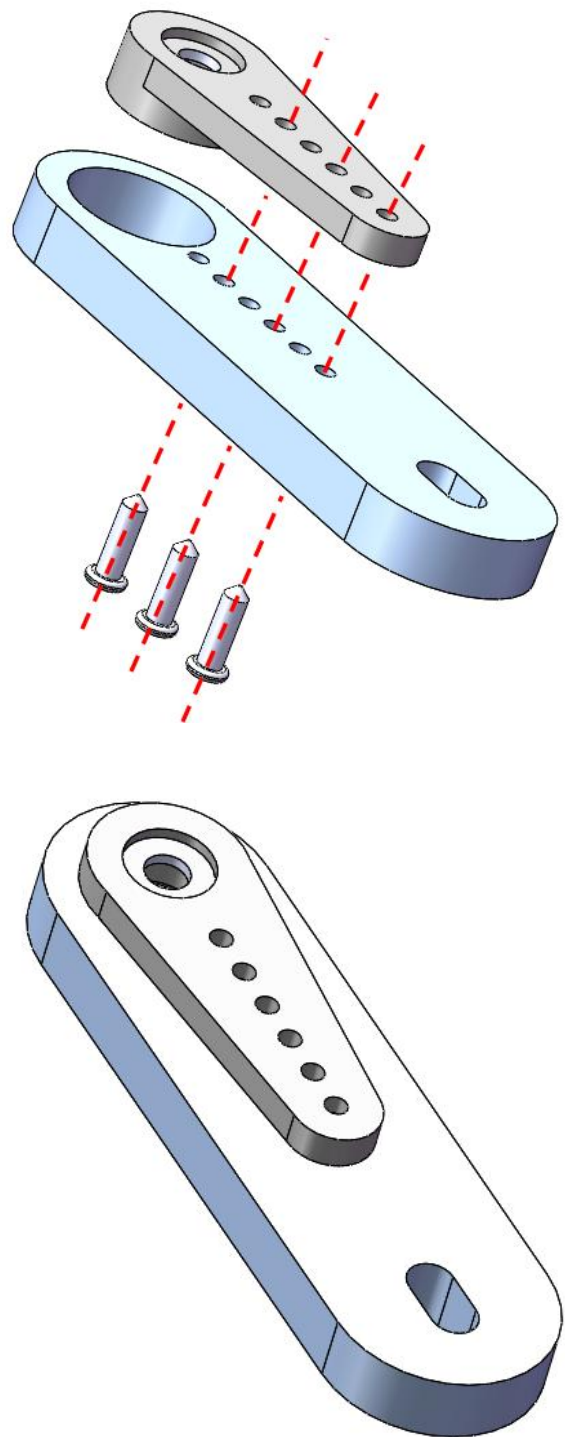


Step	6	Install the front wheel rotation axis	Tool	M3 Phillips screwdriver/M1.5 screwdriver/cross sleeve	
					
List	Part Name	Quantity	Unit	Picture	
	Step 5 part	1	PCS		

	front wheel rotation axis	1	PCS	
	Servo rotating shaft connecting rod	1	PCS	
	Servo arm (built-in servo motor)	1	PCS	
	M2.5*4 mm round head screw (Built-in servo motor)	1	PCS	
	P1.2*5 mm self-tapping screws	3	PCS	
	M2*12 mm round head screw	1	PCS	
	M2 self-locking nut	1	PCS	
	Description			Demo

A

1. Use 3 pcs M1.2*5mm self-tapping screws to install the servo arm into the servo rotating shaft connecting rod;

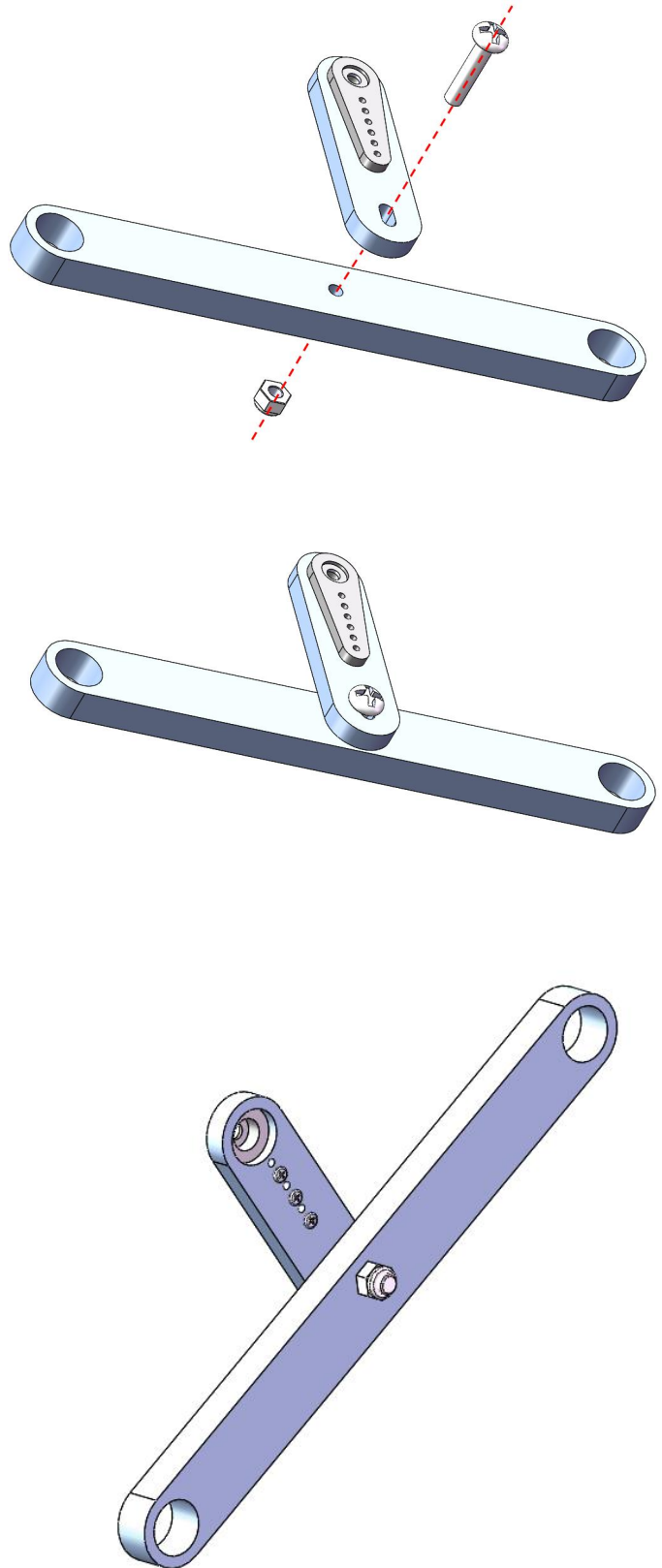


2. Use 1 pcs M2*12mm round head screw and 1 pcs M2 self-locking nut to fix the servo rotating shaft connecting rod on the front wheel rotation axis (M2 self-locking nut can not be locked, so as to ensure smooth manual rotation between the servo rotating shaft connecting rod and front wheel rotation axis);

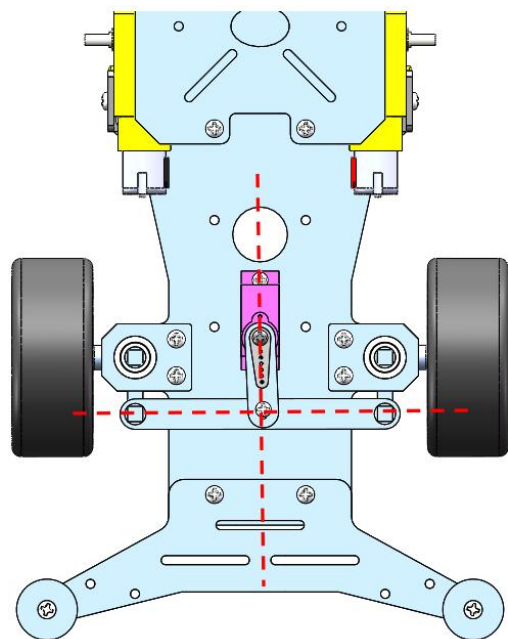
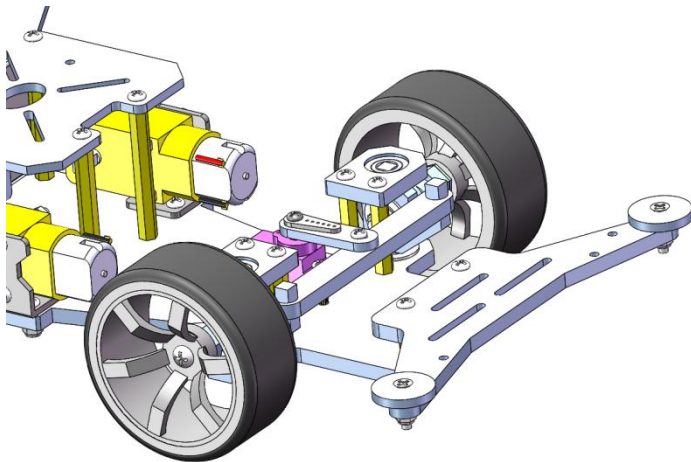
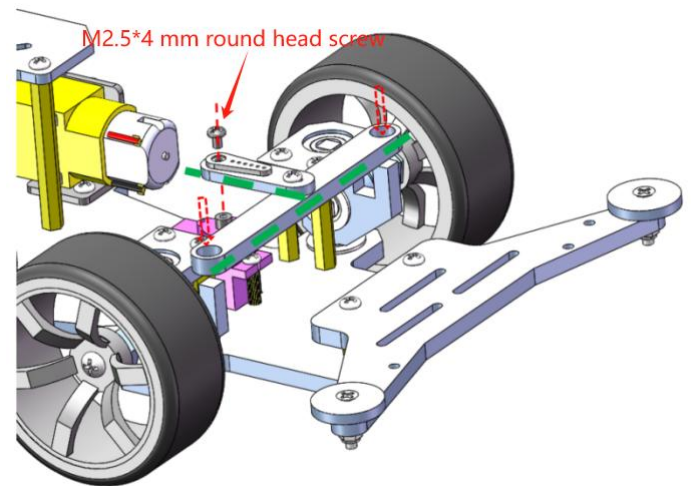
3. Use the code [Servo_90_ADJ](#) to reset the MG90 servo to 90 degrees, please refer to the Lesson 2

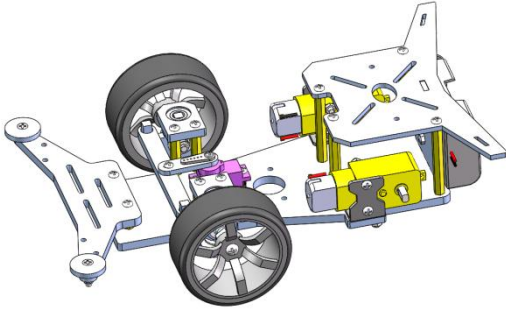
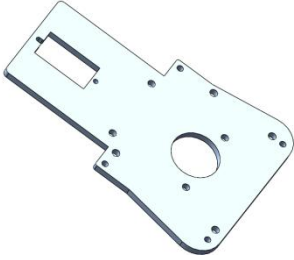
The sketch is in the path:

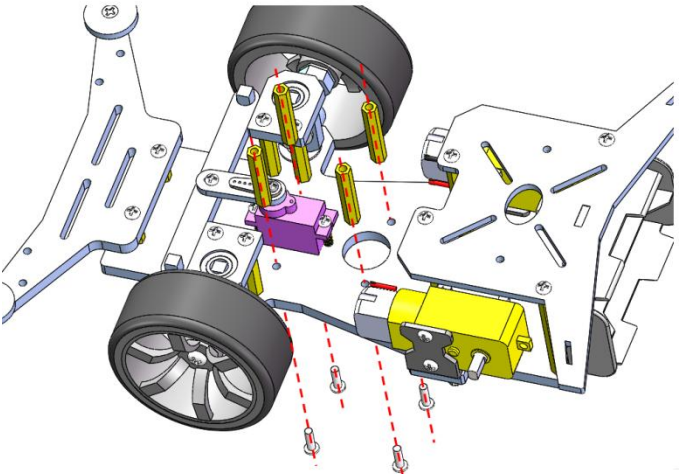
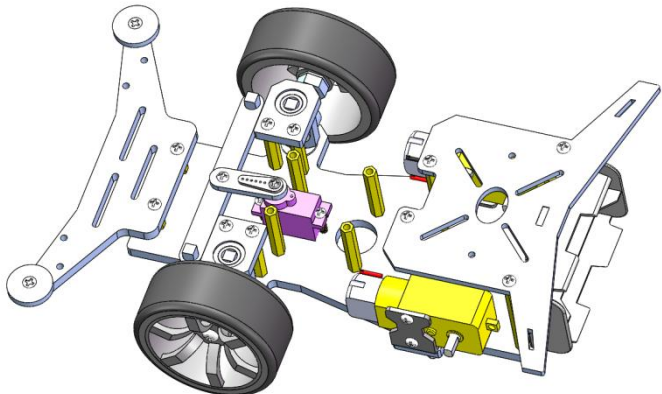
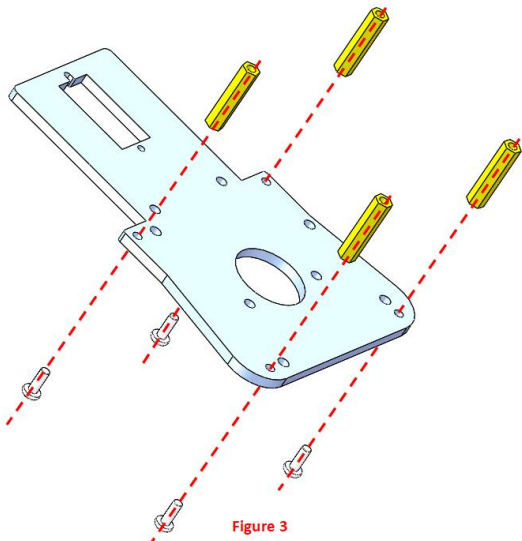
[E:\CKK0020-main\Raspberry Pi Tutorial\Code\Servo_90_ADJ](#)



4. When the servo rotating shaft connecting rod and front wheel rotation axis are kept perpendicular, install the servo rotating shaft connecting rod on the servo shaft, and the holes at both ends of front wheel rotation axis are respectively installed with the front wheel Y-shaped connecting shaft on the left and right front wheels; secure the servo arm to the servo shaft with M2.5*4mm round head screws.



Step	7	Assemble the acrylic support plate	Tool		M3 Phillips screwdriver
					
List	Part Name	Quantity	Unit	Picture	
	Step 6 part	1	PCS		
	M3*30mm double copper column	4	PCS		
	M3*10mm round head screw	4	PCS		
	M2.5*20mm double copper column	4	PCS		
	M2.5*8mm round head screw	4	PCS		
	acrylic support plate	1	PCS		

	Description	Demo
A	1. Use 4 pcs M3*10mm round head screws to fix 4 pcs M3*30mm double through copper columns on the acrylic chassis frame;	 
	2. Use 4 pcs M2.5*8mm round head screws to fix 4 pcs M2.5*20mm double through copper columns on the acrylic support plate (pay special attention to the installation direction of the double through copper columns on the acrylic support plate, refer to Figure 3 and Figure 4)	

3. Use 4 pcs M3*10mm round head screws to fix the acrylic support plate on the 4 pcs M3*30mm double copper columns which on the acrylic chassis framet;

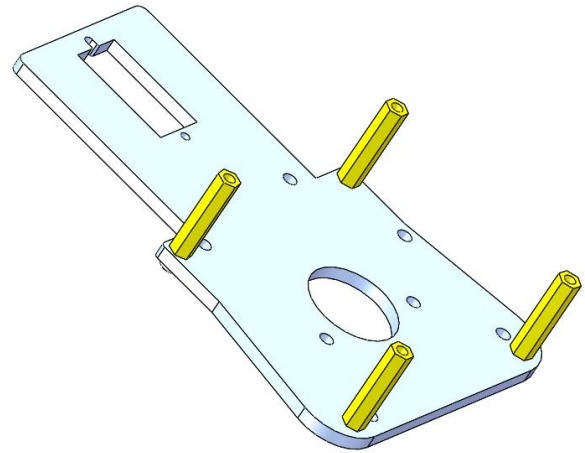
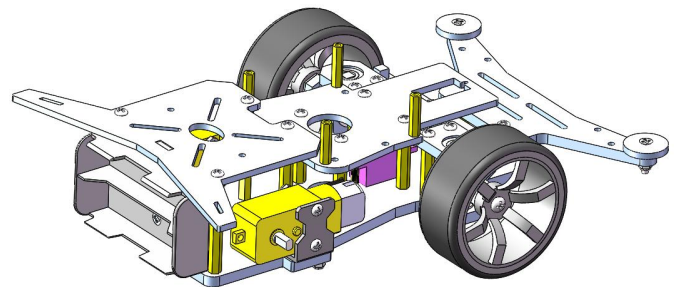
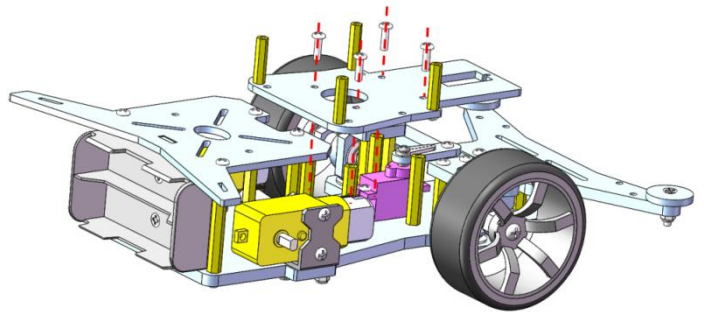
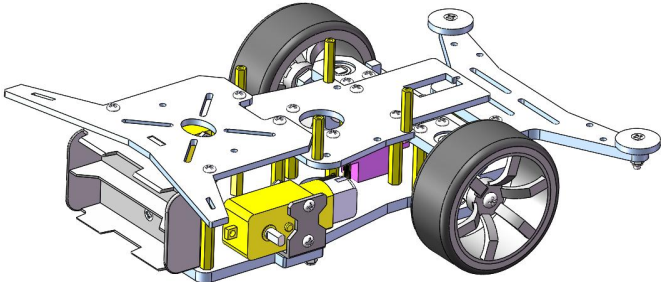
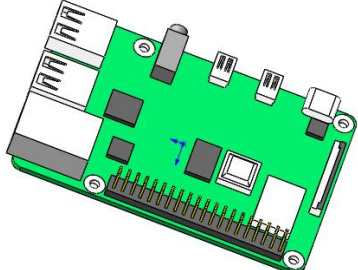
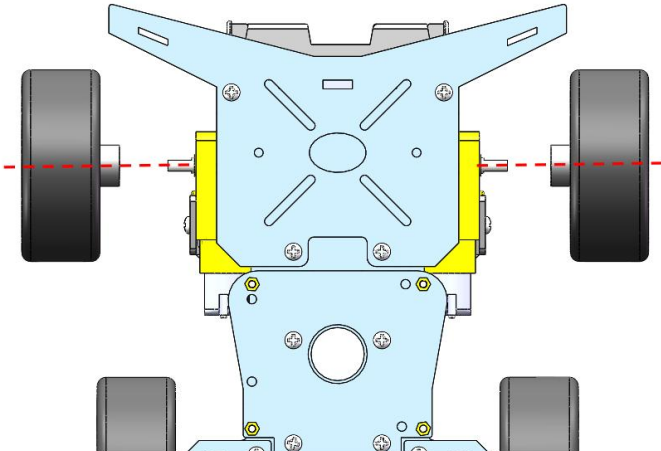


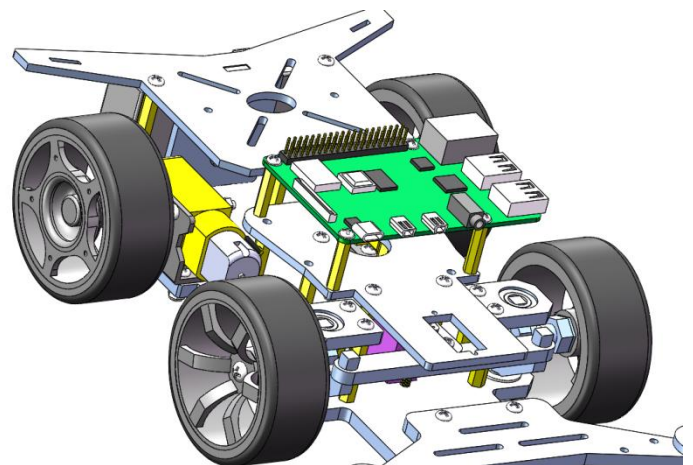
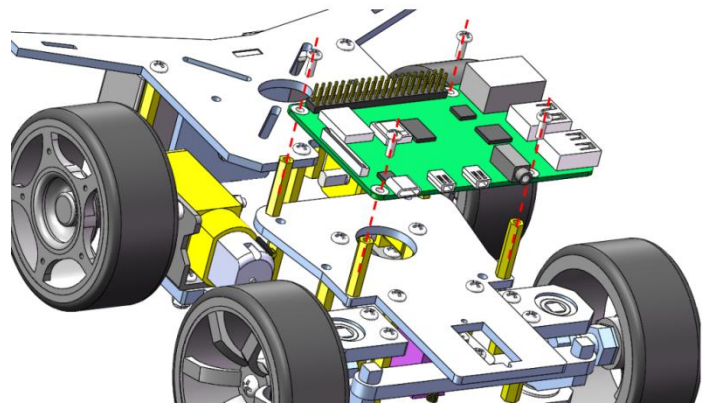
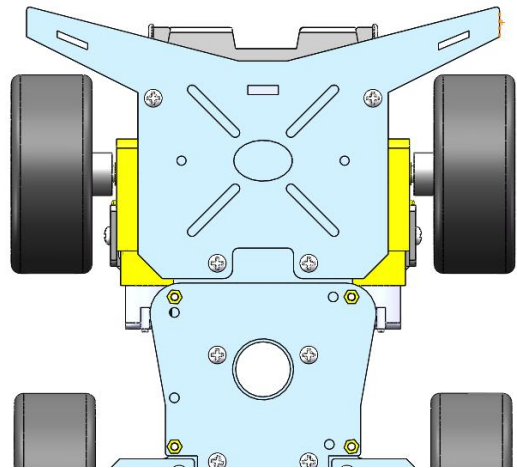
Figure 4



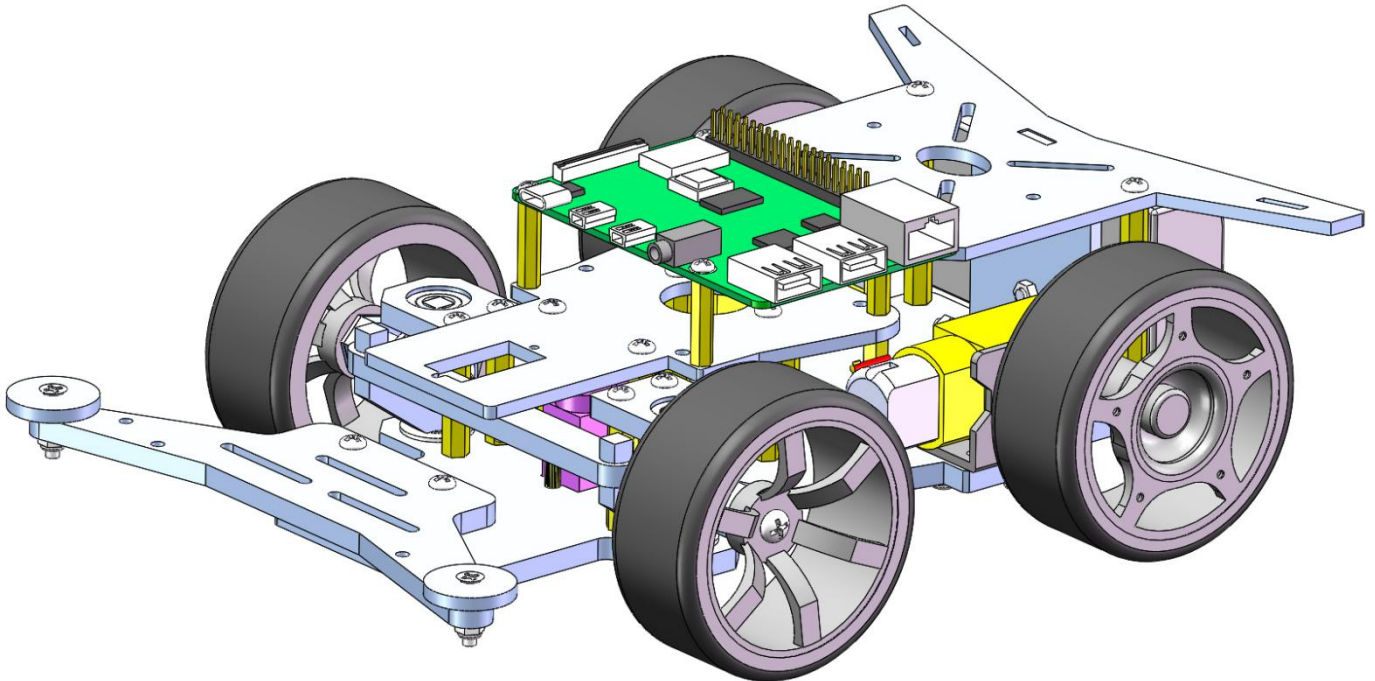
Step	8	Assemble TT motor wheel and Raspberr y Pi	Unit	Tool	M3 Phillips screwdriver
					
	Part Name	Quantity	Unit	Picture	

List	Step 7 part	1	PCS	
	TT motor wheel	2	PCS	
	Raspberry pi board (you need to prepare it yourself)	1	PCS	
	M2.5*8mm round head screw	4	PCS	
	Description			Demo
A	1. Install the two TT motor wheels on the TT motor shaft respectively;			

2. Install the raspberry pi on the acrylic support plate with 4 pcs M2.5*8mm round head screws;



Congratulations, a Real-Wheel Drive Steering Car Chassis for Raspberry pi has been installed!



3. Make your suggestion and get support

THANK YOU for participating in this assemble document!

If you find errors, omissions or you have suggestions and/or questions about this document, please feel free to contact us: cokoino@outlook.com

We will make every effort to make changes and correct errors as soon as feasibly possible and publish a revised version.

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